Nazim Belabbaci

978-421-5623 | bellabaci.nazim@gmail.com | **website**: https://nazimbl.github.io/nazim-hub/

Summary

AI research scientist (PhD, Computer Science) and Postdoctoral Research Fellow at BIDMC – Harvard Medical School. I build multi-omics pipelines and apply state-of-the-art AI to complex biomedical data, from genomic sequences to physiological signals, to advance infectious disease understanding and treatment, early cancer detection, and personalized health monitoring. Reviewer for Future Generation Computer Systems, Results in Engineering, and Digital Health.

Research & Academic Experience

Postdoctoral Research Fellow, *Beth Israel Deaconess Medical Center (BIDMC)*, Boston, MA | 2026–Present

◊ Develop AI and multi-omics integration methods for an NIH-supported HIV research program, combining single-cell RNA, chromatin, and viral integration-site data to characterize host–virus interactions.

◊ Lead the AI modeling effort in a multidisciplinary influenza vaccine project, developing deep-learning models of antigenic evolution to guide vaccine strain selection.

Graduate Research Assistant, *University of Massachusetts Lowell*, Lowell, MA | 2022–2026

◊ Led the development of an AI-enabled optical wearable for hydration monitoring, including authoring the IRB protocol, collecting and curating physiological datasets, developing machine-learning models, deploying on-device inference pipelines, and validating system against saliva ground-truth measurements.
Formal invention disclosure filed with the UML's Office of Technology Commercialization.

◊ Developed a class-conditional latent diffusion model that generates synthetic FTIR spectroscopy data to address data scarcity and class imbalance in blood-based endometrial cancer detection. Augmenting training data with synthetic spectra improved accuracy and AUC by 4% and specificity by more than 12%, outperforming SMOTE and GAN-based balancing (PhD dissertation, 2026).

◊ Developed multimodal machine-learning models for fatigue, emotion, and activity recognition using physiological and motion signals (heart rate, respiration, EDA, temperature, accelerometry), achieving up to 15 percentage-point accuracy improvements over published baselines.

PhD Student Researcher, *University of Massachusetts Lowell*, Lowell, MA | Summer 2025

◊ Lead Researcher on an NSF-funded initiative developing end-to-end AI pipelines for real-time lead detection in water supplies, addressing critical public health and environmental safety risks through analytically validated (AAS-grounded) methodologies.

PhD Research Intern, *Beth Israel Deaconess Medical Center (BIDMC)*, Boston, MA | Summer 2025

◊Built a reproducible pipeline to analyze large RNA-sequencing datasets from multiple viruses (HIV, SARS-CoV-2, influenza, and mpox), automating statistical filtering and downstream functional analysis to identify virus-associated alternative splicing.

◊Extracted COVID symptoms from unstructured clinical notes using zero-shot classification (BART), then linked symptoms to transcriptomic pathway activity scores to quantify symptom–pathway relationships in COVID. Contributed to a paper published in *Nature Immunology*.

Graduate Teaching Assistant, *University of Massachusetts Lowell*, Lowell, MA | 2023–2026

◊ Teaching AI and Assembly Language; leading weekly laboratories and recitations for 100+ students.

Graduate Research Assistant, *University of North Dakota,* Grand Forks, ND | 2021–2022

◊ Invited to join NSF-funded research supporting AI-driven drug discovery, generating large-scale molecular simulation datasets using GROMACS on HPC systems, with abstract presented at the APS March Meeting 2022.

Graduate Research Assistant, *University of North Dakota,* Grand Forks, ND | Summer 2022

◊ Developed interpretable machine-learning models for non-invasive cancer detection using cell-free DNA, contributing to a peer-reviewed publication in Communications Biology.

◊ Co-developed MotifXplorer, a web-based platform for visualizing AI predictions and model interpretability, enabling researchers to link transcription-factor binding motifs to disease-relevant genomic outcomes.

Education

PhD, Computer Science, University of Massachusetts Lowell, Lowell, MA | May 2026
Master Computer Science, University of Massachusetts Lowell, Lowell, MA | 2024
Master Computer Engineering, National Institute of E&E Engineering, Boumerdès, DZ | 2019
Bachelor Electrical & Electronic Engineering, National Institute of E&E Engineering, Boumerdès, DZ | 2017

Industry Experience

Co-Founder & Embedded Systems Engineer, *Deadline Technologies*, Bouira, DZ | 2019–2021

◊ Designed Wi-Fi/BLE-enabled embedded systems for sensor data acquisition and telemetry, managing the full engineering lifecycle from PCB design and firmware development to deployment.

Embedded Software Engineer, *EURL Microtechnologies Lab*, Boumerdès, DZ | 2018–2019

◊ Developed firmware and software for laboratory-grade electrical measurement instruments used in national educational institutions, contributing to design specification, validation, and deployment.

Awards & Honors

Recipient, LaTorre Family Scholarship Endowment Fund (UML) | 2023.
Finalist of the UND **Middleton School of Entrepreneurship Business Plan Competition** | 2022.

Winner of the UND Big Idea Challenge | 2022

◊ Awarded top honors for an AI-based early COVID-19 detection system, evaluated by industry experts through a comprehensive business and technical review.

Winner of Leapfrog International Hackathon of Algiers | 2018

◊Led a 48-hour team sprint that delivered a multi-sensor health monitoring prototype and real-time dashboard. Programmed a micro-controller to interface with custom circuitry as well as a mobile app.

Civic Engagement

Chair, Student Initiative, *AAASTE*, Pembroke Pines, FL | 2022-2024

◊ Led a student initiative organizing science, technology, and entrepreneurship webinars, coordinating a team of 20 students to promote education and research engagement within the Algerian American science community.

President & Head Project Manager, *Wameedh Scientific Club*, Boumerdès, DZ | 2016-2019

◊ Led workshops and outreach programs, supporting 100+ students in developing technical skills. Secured funding, resources, and institutional partnerships to sustain and expand the club's scientific initiatives; winner of Algeria's National Scientific Club Contest (2018).